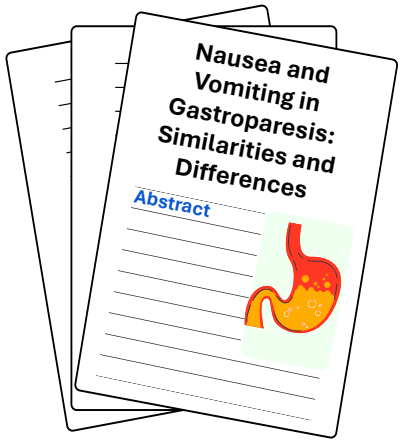


1

Biomedical Papers

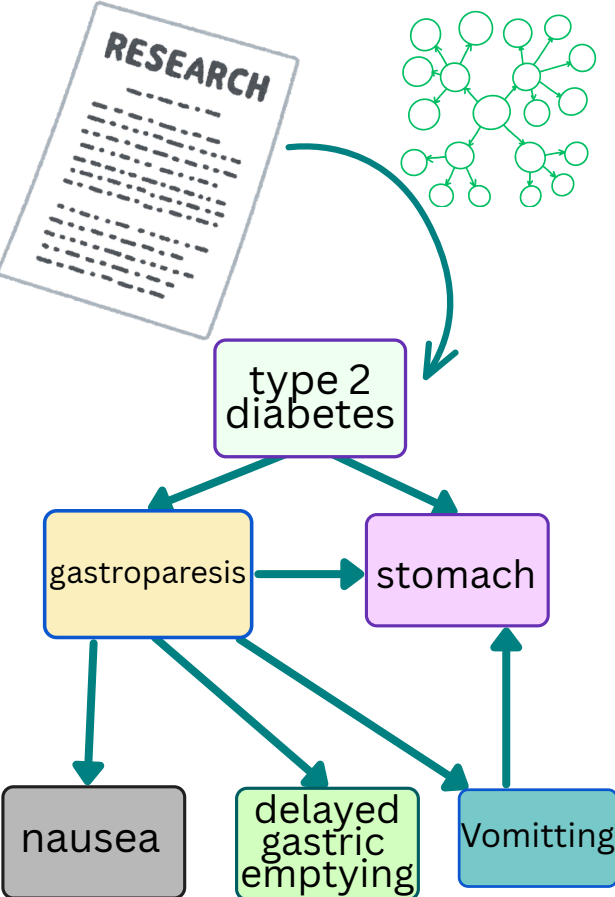


PubMed Central

2

KG Extraction

GraphMERT

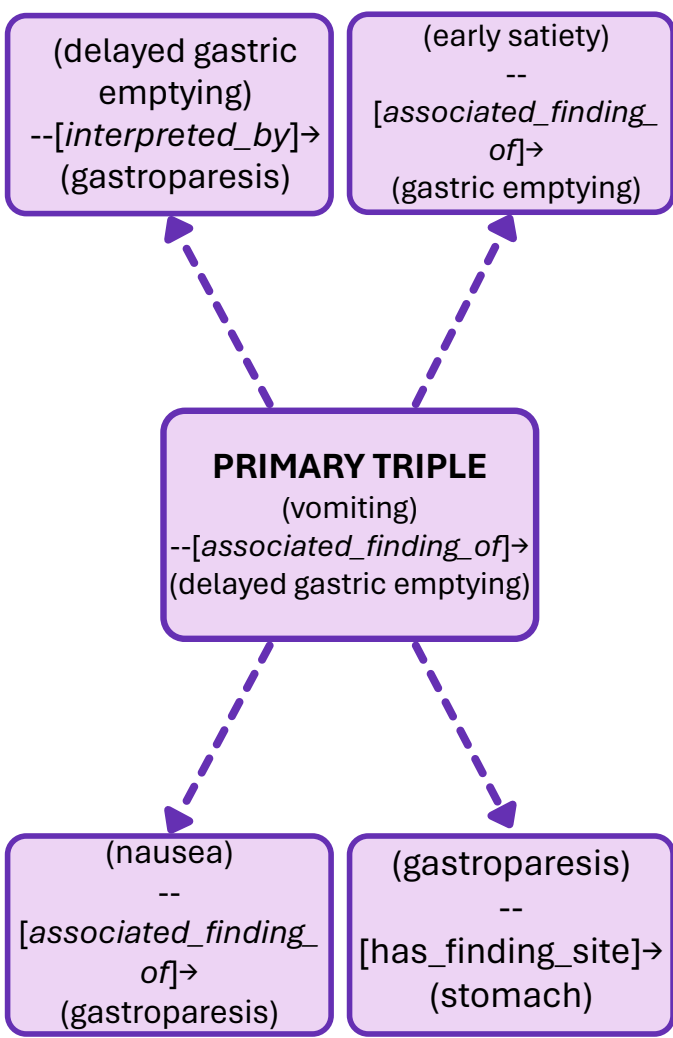


Structured Triples :
(head, relation, tail)

- (type 2 diabetes, associated_with, gastroparesis)
- (gastroparesis, associated_finding_of, nausea)
- (gastroparesis, associated_finding_of, vomiting)
- (delayed gastric emptying, affects_site, stomach)

3

Context Augmentation



primary triple
+
supporting triples

4

QA Generation

KG-grounded QA

Q: Vomiting is associated with which finding?

A. Headache
B. Delayed gastric emptying
C. Type 2 diabetes
D. Rash

Answer: B

PRIMARY TRIPLE:
(vomiting) --[associated_finding_of]→ (delayed gastric emptying)

CG-grounded QA

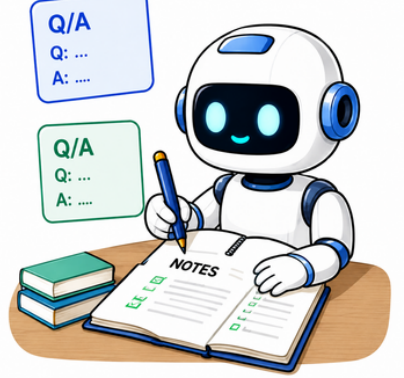
Q: A patient with type 2 diabetes has gastroparesis with nausea and vomiting. Which stomach-related finding best explains this?

A. Liver inflammation
B. Lung obstruction
C. Delayed gastric emptying
D. Kidney failure

Answer: C

5

SFT



Supervised Fine Tuning on KG- and CG-grounded QA

6

Adaptive Repair



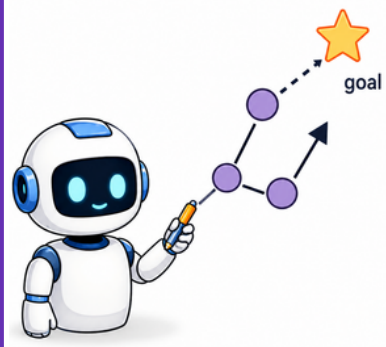
Missing Knowledge

Before: (vomiting) --[associated_finding_of]→ (delayed gastric emptying) X

After: (vomiting) --[associated_finding_of]→ (delayed gastric emptying) ✓

7

RL



Reinforcement Learning

8

Improved Models

